

Route Enumeration vs. Branching: Trade-off

Aspect	Route Enumeration	Traditional Branching
Column pool	Enumerate all routes (exponential, but bounded)	Generate on-the-fly (column generation)
Pricing sub-problem	Not needed after enum.	SPPRC at every node
Node work	Solve LP / IP directly	LP + pricing loop
Suitable for	Small/medium n (exact routes known)	Large n (column gen. scales)
Memory	High (store all routes)	Low (generate as needed)
Key algorithm	Baldacci, Mingozzi, Martello (2008)	Fukasawa, Poggi, Uchoa (2006)